ELEC 221 Lecture 14
Analysis of CT systems based on first- and second-order differential equations

Tuesday 27 October 2026

Announcements

- Quiz 7 today
- Midterm Monday 2 Nov during tutorial

Last time

We explored time-frequency duality through multiplication and convolution properties of the Fourier transform.

Last time

We expressed spectra in the magnitude phase representation

For LTI system with frequency response $H(j\omega)$ and input $x(t)$,

$|H(j\omega)|$ is the gain and $\sphericalangle H(j\omega)$ is the phase shift.

Last time

We introduced Bode plots.

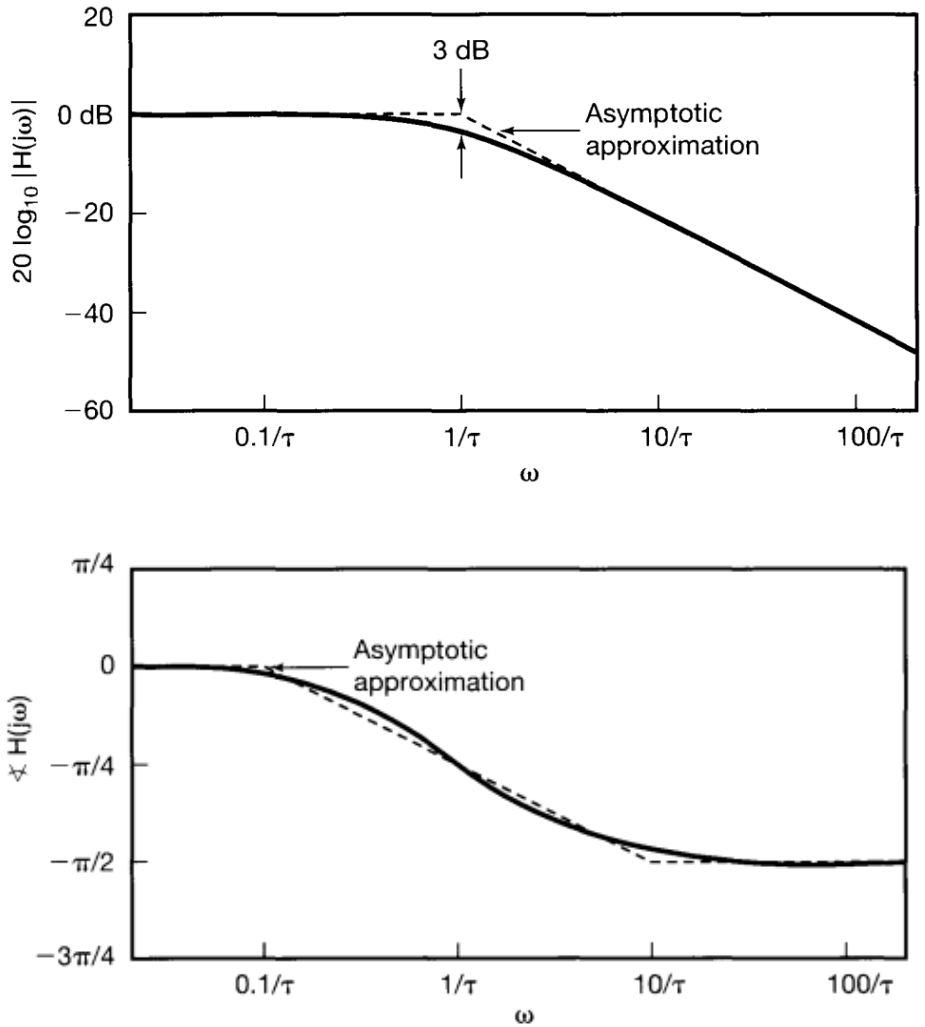


Image credit: Oppenheim 6.5

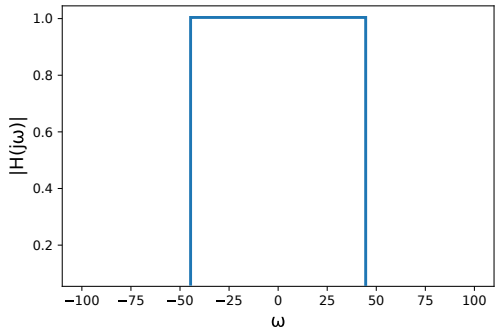
Today

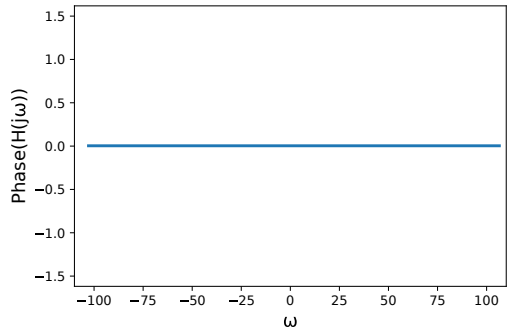
Learning outcomes:

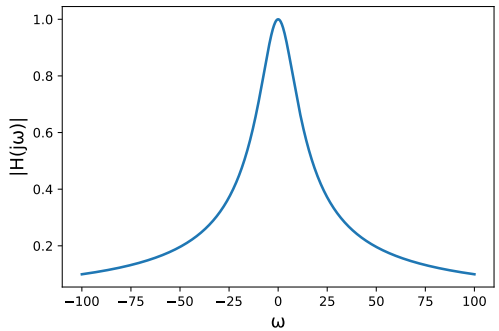
- define and compute the unit step response of a first-order system
- define the behaviour of the FT under differentiation and integration
- compute the spectrum of systems described by ODEs
- characterize the oscillatory behaviour of CT systems described by second-order differential equations

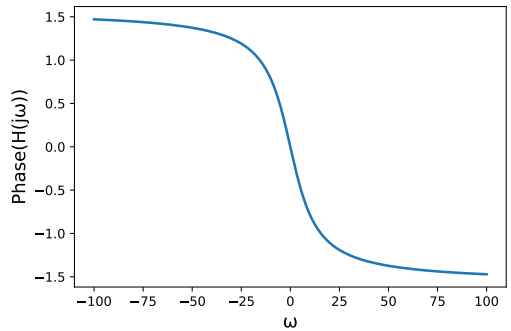
Lowpass filters

We've seen two versions of lowpass filters:









What's the difference?

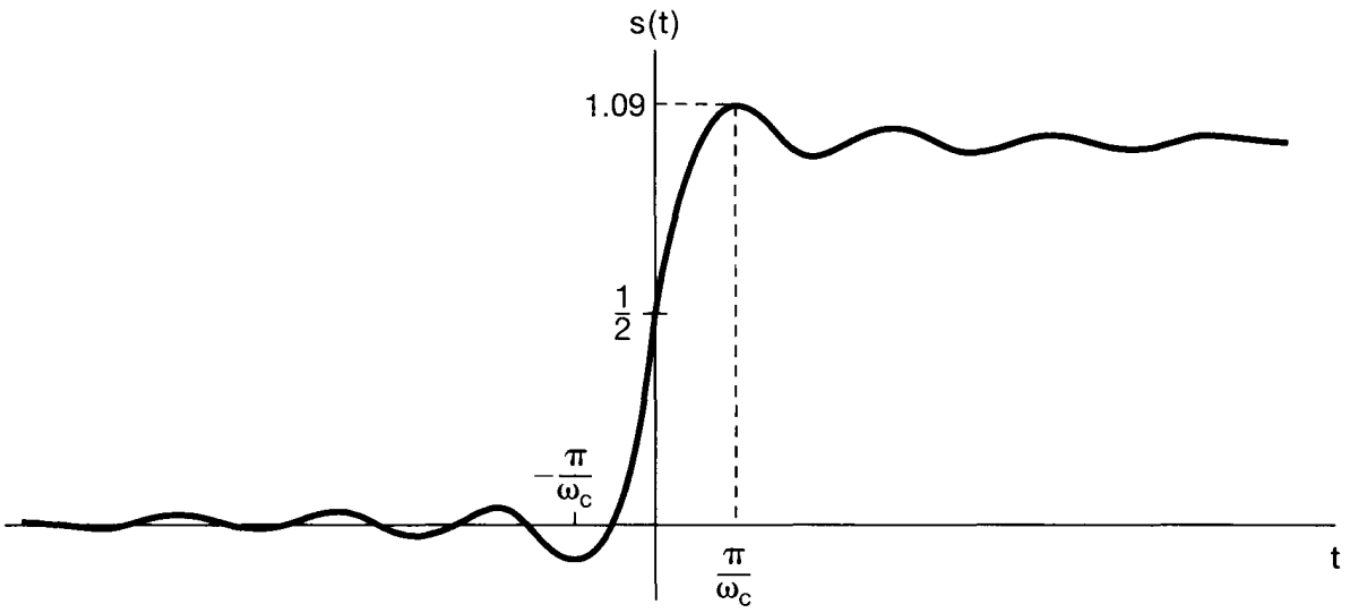
Lowpass filters

Look at their impulse responses:

Step response

It is also important to consider the *step response* of filters:

Ideal filter step response



An ideal filter leads to **ringing** in the step response.

Image credit: Oppenheim 6.3

Ideal filter step response

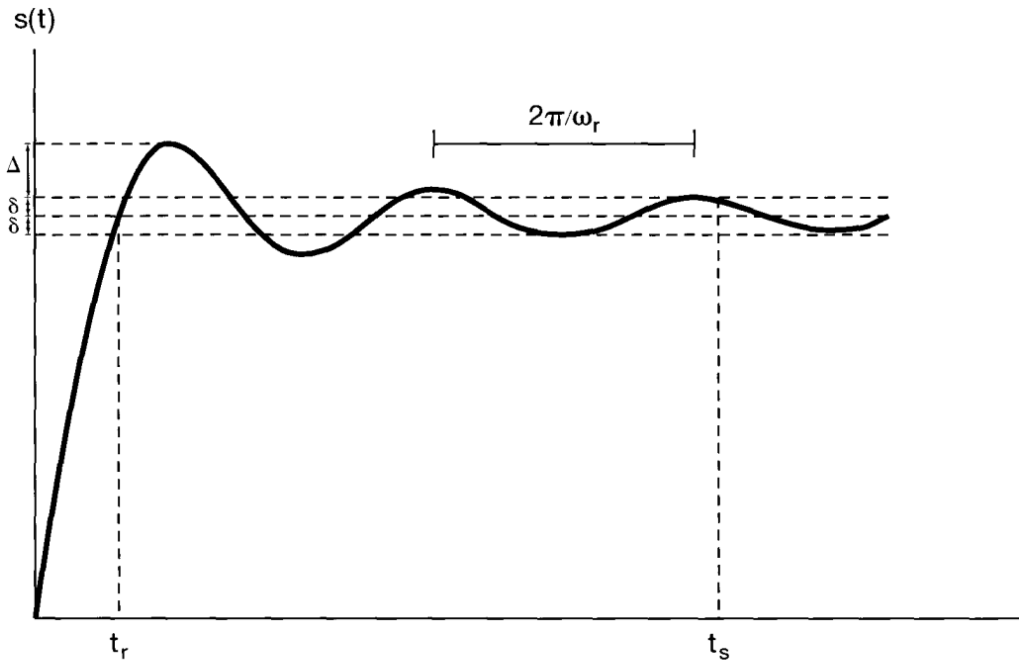


Figure 6.17 Step response of a continuous-time lowpass filter, indicating the rise time t_r, overshoot Δ, ringing frequency ω_r, and settling time t_s—i.e., the time at which the step response settles to within $\pm\delta$ of its final value.

Image credit: Oppenheim 6.4

Non-ideal filters

There are **tradeoffs** in filter design. Compromises in the frequency domain can lead to nicer behaviour in the time domain.

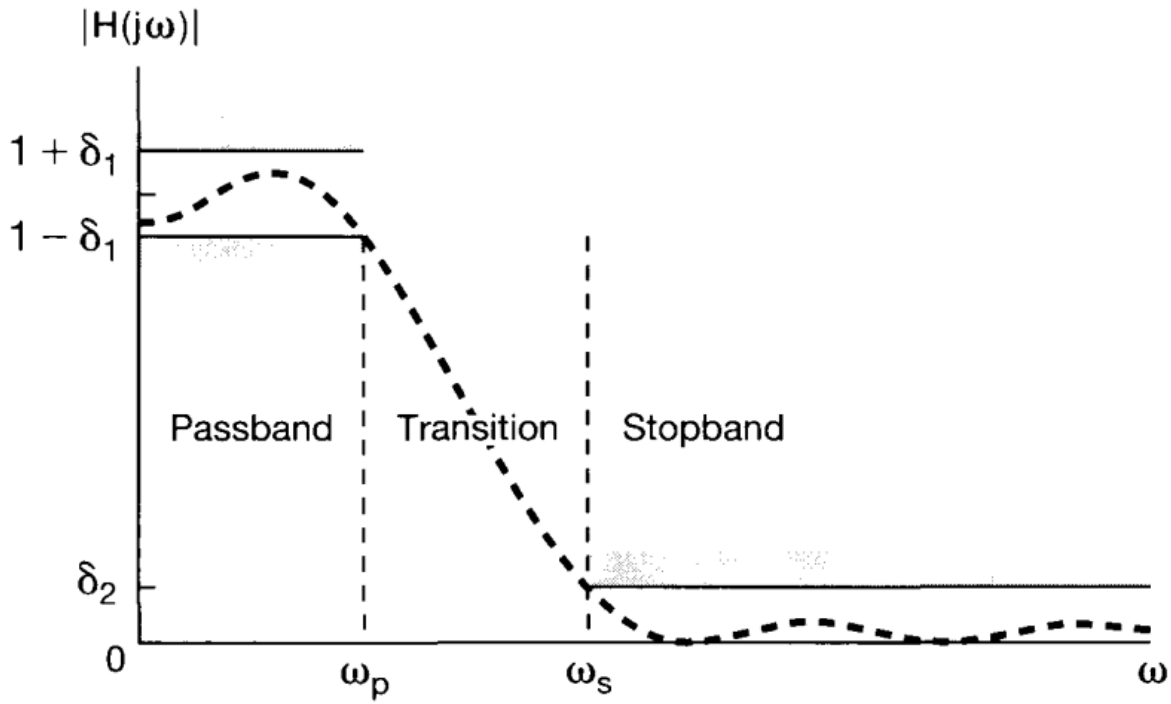


Image credit: Oppenheim 6.4

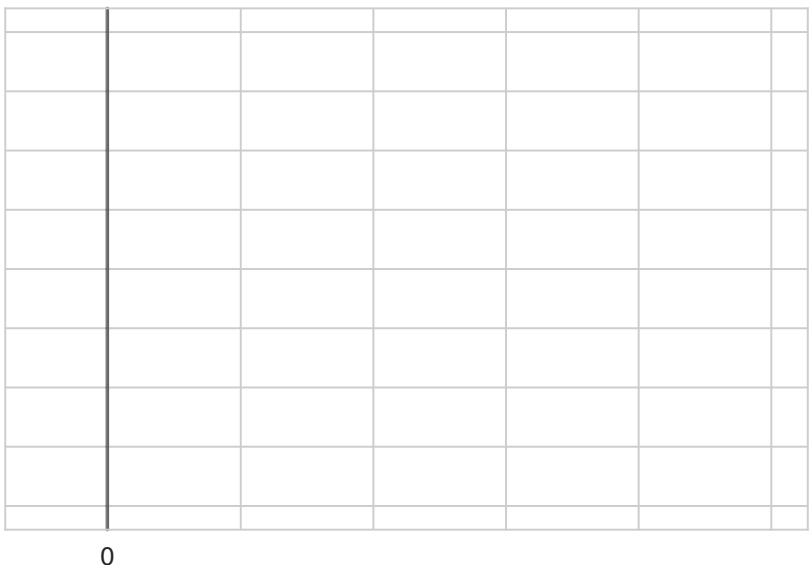
First-order systems

Exercise: What is the step response of a system described by a first-order ODE?

First-order systems

The impulse and step response of the system are

T is the **time constant** of the system.



Fourier transforms: differentiation

Consider the inverse Fourier transform:

What happens when we differentiate $x(t)$?

This means:

Fourier transforms: integration

What should happen here?

Good initial guess:

More precisely:

Fourier transforms: integration

Exercise: what are the Fourier transforms of the unit impulse and unit step?

Example: Fourier transform properties and differentiation

We can take advantage of differentiation and integration properties to simplify computations.

Suppose

What is the Fourier transform of

Two properties to take advantage of here:

Example: Fourier transform properties and differentiation

$$z(t) = \frac{d^2}{dt^2} x(t-1)$$

First, consider: $p(t) = x(t-1)$:

Fourier transforms and systems described by differential equations

Recall the LTI system described by a first-order ODE:

LTI systems can also be described by higher-order ODEs:

We will derive a general expression for the frequency response of such systems.

Fourier transforms and systems described by differential equations

$$\sum_{k=0}^{N} \alpha_k \mathcal{F}\left(\frac{d^k y(t)}{dt^k}\right) = \sum_{k=0}^{M} \beta_k \mathcal{F}\left(\frac{d^k x(t)}{dt^k}\right)$$

Fourier transforms and systems described by differential equations

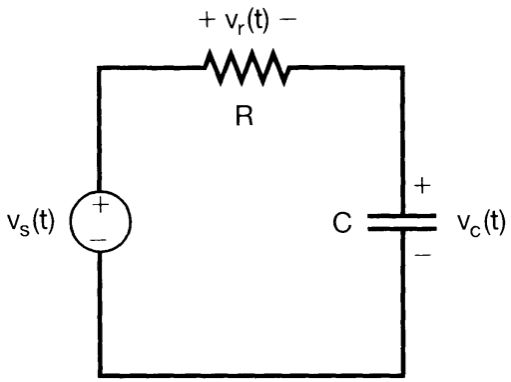
$$Y(j\omega)\sum_{k=0}^{N}\alpha_k(j\omega)^k = X(j\omega)\sum_{k=0}^{M}\beta_k(j\omega)^k$$

Final property:

This representation allows us to write down frequency response of systems described by ODEs **by inspection**! (and vice versa)

Exercise: frequency response of systems described by ODEs

What are the **impulse response** and **frequency response** of our RC circuit filter?



Exercise: frequency response of systems described by ODEs

What are the **impulse response** and **frequency response** of the system described by

Start with frequency response:

Example: frequency response of systems described by ODEs

We can now leverage this to determine the impulse response:

$$H(j\omega) = \frac{3(j\omega)^2 + 1}{(j\omega)^3 - 4j\omega}$$

Use partial fractions:

Example: frequency response of systems described by ODEs

Details are left as an exercise:

To get the impulse response, we can take the inverse Fourier transform, and leverage linearity:

Example: frequency response of systems described by ODEs

$$h(t) = -\frac{1}{4}\mathcal{F}^{-1}\left(\frac{1}{j\omega}\right) + \frac{13}{8}\mathcal{F}^{-1}\left(\frac{1}{j\omega+2}\right) + \frac{13}{8}\mathcal{F}^{-1}\left(\frac{1}{j\omega-2}\right)$$

Check Table 4.2 - two expressions to leverage:

So we have:

Second-order systems

Consider a system described by the ODE

$$\frac{d^2y(t)}{dt^2} + 2\zeta\omega_n\frac{dy(t)}{dt} + \omega_n^2 y(t) = \omega_n^2 x(t)$$

Exercise: what is the frequency response?

Second-order systems

Let's explore this in a little more detail and compute the impulse and step response of this system.

where

Three cases to consider:

- $\zeta = 1$
- $\zeta > 1$
- $\zeta < 1$

Second-order systems

$$H(j\omega) = \frac{\omega_n^2}{(j\omega - c_+)(j\omega - c_-)}, \quad c_\pm = -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2 - 1}$$

Case: $\zeta = 1$.

Use handy table of Fourier transform pairs to find

Second-order systems

$$H(j\omega) = \frac{\omega_n^2}{(j\omega - c_+)(j\omega - c_-)}, \quad c_\pm = -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2 - 1}$$

Case: $\zeta \neq 1$. Do a partial fraction expansion:

Use table of Fourier transform pairs to find

Second-order systems

$$h(t) = \frac{\omega_n}{2\sqrt{\zeta^2 - 1}} \left(e^{c_+ t} - e^{c_- t}\right) u(t), \qquad c_\pm = -\zeta\omega_n \pm \omega_n \sqrt{\zeta^2 - 1}$$

Exercise: derive the step response,

$$s(t) = \left[1 + \frac{\omega_n}{2\sqrt{\zeta^2 - 1}} \left(\frac{e^{c_+ t}}{c_+} - \frac{e^{c_- t}}{c_-}\right)\right] u(t)$$

The form of the exponential depends whether $\zeta > 1$ or $\zeta < 1$

- $\zeta < 1$: $c_\pm$ are imaginary; complex exponentials, so the response will oscillate!
- $\zeta > 1$: $c_\pm$ real and negative; decaying exponentials

Let's go plot these.

For next time

Content:

- Brief summary of discrete-time Fourier transform

Action items:

1. Study for midterm

Recommended reading:

- From today's class: Oppenheim 6.1-6.5, 6.7
- Suggested problems: 4.33-4.36, 6.1, 6.3a, 6.5, 6.9, 6.15, 6.21-6.23
- For next class: Oppenheim 5.0-5.7